ECE 49022: Electrical Engineering Senior Design Projects
Spring 2026
CRN:14362

ECE 49022 Staff

January 13, 2026

Course Information

Requisites

Senior standing and must have either completed all core courses **OR** may take your final 30000-level core course concurrently.

Course Description

Lecture sessions provide the student with background information on the design and management of projects. Formal lectures cover, for example, design for manufacturability, design for quality, test and evaluation, reliability and ethics, patents and copyrights, plus case studies. During the laboratory sessions, the students work in teams on a challenging open-ended electrical engineering project that draws on previous course work. Projects routinely involve standard design facets (such as consideration of alternative solutions, feasibility considerations and detailed system descriptions) and include a number of realistic constraints (such as cost, safety, reliability, and aesthetics).

Learning Resources, Technology, and Text

- Brightspace page
  - You can access the course via Brightspace. It is strongly suggested that you explore and become familiar not only with the site navigation, but with content and resources available for this course. See the Content tab for resources.
- Additional Readings: The list of readings is below. As we progress through the course, other resources may be added. Access to additional readings and online chapters will be through the Brightspace course readings folder and through the Library Resources Link.
  - Fundamentals of Engineering Design, 2nd Edition, Barry Hyman, Prentice hall, 2002, ISBN No. 978-0130467126.

Learning Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

1. an ability to apply engineering design to create a product that meets the specified needs of this engineering design experience with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. [2]
2. an ability to develop and conduct experimentation, analyze and interpret data, and use engineering judgment to draw conclusions related to the development of the product of this engineering design experience. [6]
3. an ability to identify, formulate, and solve complex engineering problems arising from this engineering design experience by applying principles of engineering, science, and mathematics. [1]
4. an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives associated with this design experience. [5]
5. an ability to communicate effectively with a range of audiences appropriate to this design experience in both a written report and oral presentation. [3]
6. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies to complete the engineering design experience associated with this course. [7]
7. an ability to recognize ethical and professional responsibilities associated with this engineering design experience and make informed judgments which must consider the impact of the product of this engineering design experience in global, economic, environmental, and societal contexts. [4]

Assignments

Your learning will be assessed through a combination of participation, projects, technical documentation, and a final exam spread throughout the academic period. Details on these assignments and exams, including a schedule of due dates, rubrics to guide evaluation, and guidelines on discussion participation and evaluation will be posted on the course website.

Assignment/Event	Weight
Lab Participation	5%
Technical Design	30%
Conceptual Design Proposal	5%
Manager's Meeting	10%
Midterm Demo	10%
PCB or Software Review	6%
Final Demo	15%
Final Design Review	15%
ABET Report	4%
Total:	100%

Table 1: Grade Breakdown

Lab Participation

The participation score is a discretionary part of your grade decided by your TA and Professor. Participation is graded by the TA and Professor weekly, and is defined as being physically present and contributing to project goals or other activities relevant to the course requirements. By its very nature, it is not possible to put a fully defined grading scheme on a design project without potentially punishing bad behavior and rewarding good behavior. To ensure that this class can reward students appropriately, participation is awarded for, but not limited to, the following activities:

- Helping team members
- Preparing team documentation.
- Carrying out a good plan, regardless of outcome.

Technical Design

A series of written documents, capturing the details of your technical approach and chosen design. The design document will be developed over the system as your design progresses. Some sections are "group" work, some are "individual" or sub-system specific. New sections will be completed for each submission. You are expected to continually update this document with credit in progress checks.

Solution Proposal

Initial pitch to the investors on how your team will build the project you've proposed. You may not proceed with the course until your pitch is accepted. This presentation is the primary means of doing that.You must continue to modify your design document until it is accepted by the section TA and/or Professor.You will not be allowed to turn in assignments that are due after the solution proposal until the proposal is accepted. All assignment late penalties will apply.

Manager Meeting

This is a meeting between your team and the course staff, in which you should have a complete design for your project at this point. You will be expected to explain your initial specifications, initial schematics, required parts, and sub-system interfaces. Graders will ask you open-ended questions on your current design and future plans.

Demos

Your team completes a live demonstration of the current prototype to everyone. A focus in demonstrations is in meeting product requirements and deliverables. A minimum level of deliverable can, among other things, be demonstrated by producing:

- accurate block diagram(s) that meet the major goals of the subsystem
- functional prototypes (breadboard circuits software demonstrations, etc.) of the subsystem
- accurate schematics of what has been built
- a basic understanding of the electrical and computer engineering theory behind the design

PCB Review

In this review, your TA reviews and comments on your PCB design. Your team updates the PCB design based on this feedback, and then orders, assembles, and at least sufficiently debugs the PCB.

Final Design Review

Your team gives a group presentation on the current technical status of the prototype. The group grade focuses on system-level design detail, while the individual grade focuses on your sub-system performance measurements and documentation. It is the group's responsibility to make it work. It is the individual's responsibility to test, measure, and document that performance.

ABET Report

Complete and submit answers to a set of ABET prompts. Failure to complete an ABET report will results in all team members receiving an F.

Grading

The grading scale for this course is a follows:

Letter	GPA	Percent Range
A	(4.0)	$100\% \geq x \geq 93\%$
A-	(3.7)	$93\% > x \geq 90\%$
B+	(3.4)	$90\% > x \geq 87\%$
B	(3.0)	$87\% > x \geq 83\%$
B-	(2.7)	$83\% > x \geq 80\%$
C+	(2.4)	$80\% > x \geq 77\%$
C	(2.0)	$77\% > x \geq 73\%$
C-	(1.7)	$73\% > x \geq 70\%$
D+	(1.4)	$70\% > x \geq 67\%$
D	(1.0)	$67\% > x \geq 63\%$
D-	(0.7)	$63\% > x \geq 60\%$
F	(0)	$60\% > x \geq 0\%$

Table 2: Grading Scale

In extenuating circumstances, a grade of incomplete (I) or conditional failure (E) may be awarded to a student; these determinations are made at the discretion of the instructor. Students who fall within 0.5% of a letter grade threshold may be subject to a higher letter grade; these determinations are made at the discretion of the instructor.

Penalties

Penalties related to your conduct within the course. These penalties do not include points deduction for data presented during the course.

Unexcused absence from solution proposal	100% deduction on the assignment
Unexcused absence from midterm demo	100% deduction on the assignment
Unexcused absence from final demo	100% deduction on the assignment
Unexcused absence from final design review	100% deduction on the assignment
Cheating, unethical behavior, plagiarism	100% deduction from assignment
Late submission of any document	10% + 5% per day
Complaining to the audience at any design review or demonstration	10-25% deduction from assignment
Failure to credit sources	10% deduction from assignment